

VOICECARE AI

Safe Growth Operating System

Workflow and Technical Architecture Guide

Sequential Multi-Agent Framework for Safe Brand Scaling

A technical deep dive into the 6-agent sequential pipeline engineered to accelerate LinkedIn follower growth, capture category share-of-voice, and optimize AI engine visibility (GEO/AEO/ASO) while adhering to strict healthcare compliance (PHI protection) and platform anti-spam boundaries.

Document Scope:

Prototype & Architecture Specification

Target Follower Goal:

2,900 to 10,000 Followers in 90 Days (~590 net-new/week)

Core Technologies:

HTML5, CSS3, ES Modules, Vercel Serverless, Groq API

Safety Posture:

100% Pure Research & Drafting (Human-in-the-Loop Required)

1. Executive Summary

The VoiceCare AI Safe Growth Engine is an operating console designed to scale B2B marketing and brand visibility. Traditionally, rapid social scaling relies on automated bots, scrapers, or bought leads - tactics that lead to severe platform penalties, profile bans, and brand erosion. In healthcare, these risks are compounded by the necessity to safeguard patient data and protect private health histories under strict HIPAA frameworks.

This platform acts as an administrative cockpit. It utilizes a sequential chain of six domain-specific AI sub-agents to analyze competitors, draft tailored content, audit SEO blogs, calculate growth velocities, compile third-party authority citation lists, and map assets to conversion funnels. The core architectural boundary is simple: the AI researches and drafts, but never executes external API actions. Humans review, verify, and post everything manually.

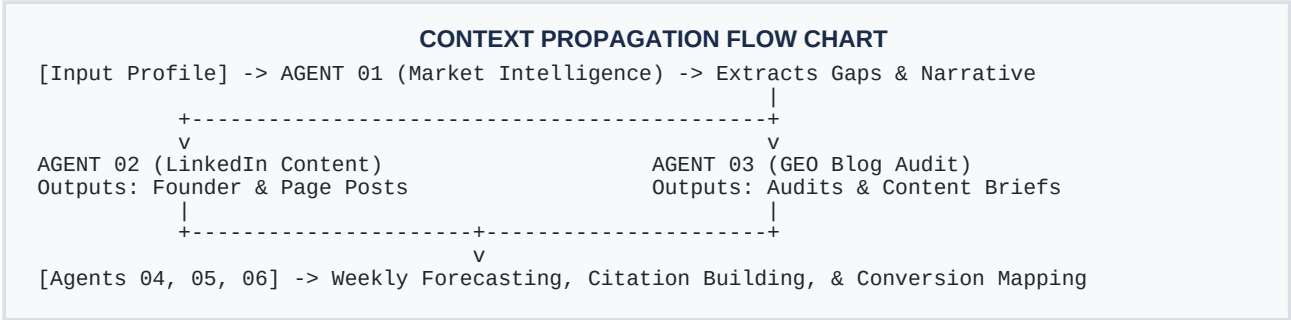
2. Technical Stack Architecture

The application is structured to minimize client-side dependencies, reduce payload weights, and securely isolate sensitive credentials. The stack components are:

- **HTML5 UI:** A single-page, responsive console layout utilizing semantic elements. The workspace is divided into a Company Profile input panel, an Orchestration Control bar (for single-agent or full sprint execution), and a grid of six agent output cards.
- **Vanilla CSS3:** A custom, responsive design system engineered using CSS Grid and Flexbox. Features built-in CSS variables for light/dark accenting, scroll-safe code blocks, and dynamic status dots (green for completed, amber for running, and red for errors) representing agent lifecycle states.
- **Vanilla JS & ES Modules:** No frontend compilation (Babel, Webpack, etc.) or JS frameworks (React, Vue) are required. Native ES Modules handle form state retrieval, coordinate DOM render updates, parse raw API responses, and run the sequential orchestration flows using async fetch handlers.
- **Vercel Serverless Functions:** All model orchestration is run through an API endpoint (/api/agent.js). The client UI never calls Groq directly, meaning the Groq API key remains safely locked inside Vercel's secure server environment, preventing browser leakage.
- **Groq Inference API:** High-velocity inference utilizing Groq LLaMA models. Key agents utilize primary models (such as groq/compound for web-search capability) and fall back to high-capacity alternatives (llama-3.3-70b-versatile or llama-3.1-8b-instant) in case of rate-limiting or service outages.

3. Sequential Multi-Agent Workflow

A key design pattern of the Safe Growth Engine is context propagation. Rather than treating each agent as an isolated query, the system runs a sequential sprint where findings from upstream intelligence agents are structured and forwarded into downstream creative and planning agents.



In a Full Sprint execution, the browser coordinates this sequence automatically:

- **Step 1: Market Analysis:** Agent 01 (Market Intelligence) executes first. It reads the company profile and competitors, compiles a competitive snapshot and a Share-of-Voice (SOV) scorecard, and identifies the Top 5 Content Gaps.
- **Step 2: Context Extraction:** The JavaScript orchestrator extracts the generated 'Top 5 Content & Category Gaps' and 'Category Narrative' sections from Agent 01's response using clean header boundary parsers.
- **Step 3: Creative & SEO Directives:** The extracted content is injected as competitive context variables into the payloads for Agent 02 (LinkedIn Content) and Agent 03 (GEO Visibility), ensuring the generated post drafts and blog briefs directly target competitor weaknesses.
- **Step 4: Continuous Pipeline:** The remaining agents (Agent 04 Growth, Agent 05 Citations, Agent 06 Conversion) consume metrics, directory lists, and source assets to complete the comprehensive growth sprint pack, which is saved as a single unified document.

4. Detailed Agent Profiles

Each agent runs under strict operational constraints defined in its system prompt configuration. The tables below specify the core parameters, system behaviors, and required markdown outputs.

Agent 01 - Market Intelligence + Share of Voice

Temp: 0.4 | Max Tokens: 2200

Models: Primary: groq/compound | Fallback: llama-3.3-70b-versatile
Behavior: Identifies market gaps, catalogs competitor positioning, and builds SOV matrices.
Outputs: *Competitor Snapshot, Competitor Scorecard, Top 5 Gaps, Differentiation Angles, Category Narrative, Safe Actions*

Agent 02 - LinkedIn Content + Founder Growth

Temp: 0.7 | Max Tokens: 2400

Models: Primary: llama-3.3-70b-versatile | Fallback: llama-3.1-8b-instant
Behavior: Drafts LinkedIn posts for company and leadership, outline slides, comments, and amplification kits.
Outputs: *Company Posts, Founder Posts, Employee Advocacy, Carousel Outline, Comments, Amplification Kit, Repurposing Plan*

Agent 03 - AI Search / GEO-AEO-ASO Audit

Temp: 0.4 | Max Tokens: 2600

Models: Primary: groq/compound | Fallback: llama-3.3-70b-versatile
Behavior: Conducts existing blog audits and structures GEO content briefs to capture AI engine citations.
Outputs: *Existing Blog Audit, Content Brief, Citation Testing Protocol, Content Gap Summary, Safe Actions*

Agent 04 - Growth Measurement + Forecasting

Temp: 0.3 | Max Tokens: 1900

Models: Primary: llama-3.3-70b-versatile | Fallback: llama-3.1-8b-instant
Behavior: Performs follower growth pace arithmetic and tracks visibility indices against goal pacing.
Outputs: *Pace-to-Goal Calculation, Reverse Follower Model, Visibility Scorecard, Metrics Observations, Next Experiments*

4. Detailed Agent Profiles (Cont.)

Agent 05 - Third-Party Citation + Authority

Temp: 0.4 | Max Tokens: 2200

Models: Primary: groq/compound | Fallback: llama-3.3-70b-versatile
Behavior: Discovers authoritative third-party directories, earned media opportunities, and compliance pages.
Outputs: Citation Opportunity List, Priority Targets, 5 Pitch Angles, Trust Asset Requirements, Safe Actions

Agent 06 - Conversion Asset + Repurposing

Temp: 0.6 | Max Tokens: 2400

Models: Primary: llama-3.3-70b-versatile | Fallback: llama-3.1-8b-instant
Behavior: Maps user attention to conversion funnels (demos/webinars) and repurposes assets into multi-channel content.
Outputs: Conversion Asset Map, CTA Mapping, Repurposing Matrix, Original Benchmark Asset Ideas, Safe Actions

5. Safety Boundaries & Compliance

To ensure corporate safety, protect user accounts, and prevent regulatory issues, a shared safety core is interpolated into the system prompt of every agent. These eight rules override any user request or agent inference behavior.

SHARED COMPLIANCE BOUNDARY RULES (SAFETY_CORE)

1. NO AUTOMATION: The agent only writes drafts; it has no access to post or edit social media.
2. NO SCRAPING: The agent only uses public information or manually provided analytics.
3. NO FABRICATION: The agent never invents names, numbers, or claims; missing details use placeholders.
4. NO AUTO-ENGAGEMENT: No browser automation, proxy networks, fake accounts, or pods.
5. ZERO PHI INGESTION: The agent never accepts or references Protected Health Information.
6. COMPLIANCE GUARDRAILS: No clinical, medical, or security claims without explicit verification.
7. ASO STANDARDS: Focus ASO entirely on AI Search Optimization / Answer Search Optimization.
8. MARKDOWN ONLY: The agent must output clean, raw Markdown structures with no meta preambles.

By enforcing these guardrails at the prompt level and maintaining a strict Human-in-the-Loop checkpoint, VoiceCare AI scales its LinkedIn reach and improves search citations while keeping its digital presence 100% compliant and threat-free.